

## LOEI, Thailand

WMO: 483530

Lat: 17.4528N

Lon: 101.7306E

Elev: 834

StdP: 14.26

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|-----|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%  |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                      | (o)  | (p)   |
| 12            | 53.2       | 56.6 | 45.2                          | 45.8 | 66.7 | 48.6 | 52.2 | 67.6 | 9.8                   | 78.4 | 9.2 | 78.7 | 0.6                      | 0    | 0.282 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 4             | 21.5                   | 100.6           | 74.0 | 98.2 | 74.4 | 95.7 | 74.8 | 80.1                | 89.3 | 79.5 | 88.6 | 79.0 | 88.0 | 5.7                  | 180  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 78.1                            | 150.7 | 84.0 | 77.3 | 146.5 | 83.0 | 76.9 | 144.4 | 82.6 | 44.7          | 89.6 | 43.9 | 89.0 | 43.3 | 88.6 | 86.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |     |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |     |
| (4) 11.2          | 9.4  | 7.8 | DB                         | 47.8 | 103.6              | 4.7 | 2.5                                                | 44.4 | 105.4      | 41.7 | 106.9      | 39.0 | 108.3      | 35.6 | 110.1 | (4) |
| (5)               |      |     | WB                         | 46.2 | 82.5               | 4.3 | 1.8                                                | 43.1 | 83.8       | 40.6 | 84.8       | 38.2 | 85.8       | 35.0 | 87.1  | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 79.9   | 73.1  | 76.9  | 81.3  | 84.5  | 83.7  | 83.6  | 82.8  | 82.1  | 81.4  | 80.0  | 77.0  | 72.7  |
|                                                                        | DBStd       | 5.24   | 4.36  | 4.77  | 4.25  | 3.65  | 3.10  | 2.25  | 2.29  | 2.18  | 2.05  | 2.59  | 3.78  | 4.83  |
|                                                                        | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | HDD65       | 12     | 4     | 1     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 6     |
|                                                                        | CDD50       | 10929  | 717   | 753   | 971   | 1036  | 1045  | 1009  | 1018  | 995   | 941   | 931   | 810   | 703   |
|                                                                        | CDD65       | 5467   | 256   | 334   | 507   | 586   | 580   | 559   | 553   | 530   | 491   | 466   | 360   | 245   |
|                                                                        | CDH74       | 54328  | 2224  | 3433  | 5697  | 6793  | 6073  | 6056  | 5736  | 5135  | 4362  | 3975  | 2920  | 1924  |
|                                                                        | CDH80       | 23027  | 825   | 1697  | 3131  | 3674  | 2692  | 2482  | 2154  | 1779  | 1452  | 1449  | 1087  | 605   |
| (14) Wind                                                              | WSAvg       | 2.4    | 2.2   | 2.4   | 2.6   | 2.6   | 2.5   | 2.5   | 2.7   | 2.6   | 2.2   | 2.0   | 2.0   | 2.1   |
| (15) Precipitation                                                     | PrecAvg     | 48.0   | 0.2   | 0.6   | 1.5   | 3.7   | 8.0   | 5.8   | 5.4   | 7.4   | 9.8   | 4.4   | 0.7   | 0.2   |
|                                                                        | PrecMax     | 60.7   | 1.7   | 2.1   | 2.6   | 7.0   | 15.3  | 13.3  | 7.7   | 13.8  | 20.8  | 8.3   | 2.1   | 1.6   |
|                                                                        | PrecMin     | 39.7   | 0.0   | 0.0   | 0.0   | 0.8   | 3.1   | 1.9   | 2.0   | 4.1   | 5.2   | 0.8   | 0.0   | 0.0   |
|                                                                        | PrecStd     | 6.1    | 0.5   | 0.7   | 0.8   | 1.9   | 3.2   | 2.9   | 1.7   | 2.9   | 4.6   | 2.4   | 0.7   | 0.4   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 93.2  | 98.4  | 102.5 | 105.1 | 100.7 | 96.8  | 96.1  | 93.7  | 93.2  | 93.1  | 93.5  | 91.4  |
|                                                                        |             | MCWB   | 69.7  | 71.2  | 72.1  | 74.6  | 77.6  | 78.1  | 77.9  | 78.1  | 78.6  | 77.6  | 74.8  | 73.1  |
|                                                                        | 2%          | DB     | 90.0  | 95.5  | 99.8  | 102.0 | 97.1  | 94.1  | 93.1  | 91.6  | 91.2  | 90.8  | 90.9  | 88.2  |
|                                                                        |             | MCWB   | 70.3  | 70.7  | 71.7  | 75.2  | 77.7  | 77.8  | 77.6  | 77.9  | 78.3  | 76.8  | 74.6  | 72.1  |
|                                                                        | 5%          | DB     | 87.7  | 93.3  | 97.4  | 98.8  | 94.3  | 92.2  | 91.1  | 89.9  | 89.4  | 89.2  | 88.6  | 86.0  |
|                                                                        |             | MCWB   | 69.8  | 71.1  | 72.1  | 75.5  | 78.0  | 77.8  | 77.3  | 77.5  | 77.9  | 76.1  | 73.8  | 71.1  |
|                                                                        | 10%         | DB     | 84.9  | 90.3  | 94.6  | 95.6  | 91.5  | 90.2  | 89.1  | 88.1  | 87.5  | 87.5  | 86.3  | 83.6  |
|                                                                        |             | MCWB   | 68.7  | 70.9  | 72.6  | 75.7  | 77.9  | 77.7  | 77.1  | 77.3  | 77.6  | 75.7  | 72.8  | 69.7  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 75.3  | 76.0  | 77.4  | 79.6  | 81.2  | 80.9  | 80.4  | 80.6  | 80.8  | 79.9  | 78.1  | 76.6  |
|                                                                        |             | MCDB   | 86.1  | 90.9  | 91.2  | 94.0  | 92.1  | 89.9  | 89.2  | 88.6  | 88.5  | 87.9  | 87.2  | 85.7  |
|                                                                        | 2%          | WB     | 73.3  | 74.6  | 76.0  | 78.3  | 79.8  | 79.8  | 79.2  | 79.5  | 79.5  | 78.7  | 76.8  | 74.6  |
|                                                                        |             | MCDB   | 83.9  | 88.9  | 90.7  | 92.3  | 90.7  | 88.8  | 87.7  | 87.4  | 87.3  | 86.8  | 85.5  | 83.6  |
|                                                                        | 5%          | WB     | 71.8  | 73.4  | 75.1  | 77.5  | 79.0  | 79.1  | 78.6  | 78.7  | 78.8  | 77.9  | 75.7  | 72.9  |
|                                                                        |             | MCDB   | 82.2  | 87.2  | 90.0  | 91.1  | 89.5  | 87.9  | 86.8  | 86.6  | 86.6  | 85.6  | 83.8  | 81.7  |
|                                                                        | 10%         | WB     | 70.3  | 72.2  | 74.2  | 76.8  | 78.3  | 78.5  | 78.0  | 78.0  | 78.2  | 77.1  | 74.6  | 71.6  |
|                                                                        |             | MCDB   | 81.1  | 85.0  | 88.3  | 90.0  | 88.1  | 86.8  | 85.7  | 85.2  | 85.2  | 84.4  | 82.6  | 80.7  |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 24.8  | 26.2  | 24.8  | 21.5  | 16.1  | 13.9  | 12.9  | 12.4  | 13.0  | 15.4  | 19.3  | 22.2  |
|                                                                        | 5% DB       | MCDBR  | 28.1  | 30.3  | 30.7  | 27.3  | 20.7  | 16.9  | 15.9  | 15.1  | 15.9  | 17.9  | 21.5  | 23.5  |
|                                                                        |             | MCWBR  | 10.5  | 9.7   | 8.4   | 6.3   | 5.4   | 4.5   | 4.6   | 4.8   | 5.4   | 5.9   | 7.4   | 8.9   |
|                                                                        | 5% WB       | MCDBR  | 23.2  | 25.9  | 24.4  | 21.8  | 17.4  | 14.8  | 13.9  | 13.8  | 14.3  | 15.8  | 17.9  | 20.5  |
|                                                                        |             | MCWBR  | 9.1   | 8.7   | 7.5   | 6.3   | 5.6   | 4.8   | 4.9   | 5.1   | 5.5   | 5.9   | 6.7   | 8.1   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.466 | 0.573 | 0.674 | 0.626 | 0.495 | 0.445 | 0.433 | 0.448 | 0.495 | 0.540 | 0.454 | 0.456 |
|                                                                        | taud        |        | 2.108 | 1.848 | 1.653 | 1.801 | 2.188 | 2.356 | 2.385 | 2.321 | 2.174 | 2.006 | 2.221 | 2.155 |
|                                                                        | Ebn at Noon |        | 258   | 236   | 217   | 229   | 258   | 268   | 272   | 270   | 258   | 241   | 260   | 257   |
|                                                                        | Edh at Noon |        | 48    | 65    | 82    | 71    | 48    | 40    | 39    | 42    | 48    | 55    | 43    | 45    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1474  | 1615  | 1630  | 1748  | 1643  | 1543  | 1445  | 1400  | 1427  | 1492  | 1533  | 1427  |
|                                                                        | RadStd      |        | 98    | 74    | 103   | 95    | 114   | 110   | 99    | 90    | 89    | 102   | 64    | 58    |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |  | +0.56 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | +7    | +234  | +230  |
| (47) Regional (17 neighbors) |  | +0.52 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +209  | +209  |

Nomenclature: See separate page